

Title of the Project:

Development of an **Intelligence Care (I-Care)** Program for Dementia Caregivers

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Abstract

Caring for people with dementia is extremely stressful for caregivers, leading to psychiatric and physical illness. Without educational resources, caregivers have difficulty managing behavioral symptoms as dementia progresses and disproportionately experience strain and burnout. Generative artificial intelligence (GenAI), particularly large language models (LLMs) such as ChatGPT, have demonstrated potential for delivering personalized, on-demand education and psychosocial support. This project will develop the **Intelligence Care (I-Care)** program, a GPT based digital companion that integrated validated educational materials from Rutgers Health's Care2Caregivers program. The I-Care program will [integrate multi-agent conversational intelligence, retrieval-augmented education, and peer-support modules to deliver personalized, evidence-based guidance for dementia caregivers. Through dynamic dialogues, contextual retrieval of Rutgers Health's Care2Caregivers materials, and virtual peer interactions, I-Care aims](#) to improve dementia caregiver's knowledge, coping skills, and well-being in New Jersey. This study will proceed in three steps. Step 1 is to design the program curriculum and conversational scenarios with expert and Care2Caregivers program. Step 2 is to develop the **I-Care** prototype using GPT-based natural-language understanding and adaptive learning algorithms, and Step 3 is to pilot testing to assess usability and refine the system. Finding of this study will support large scale testing and integration into Rutgers Health's statewide dementia caregiver support services.

This grant will be used as leverage to obtain future funding to test the feasibility and efficacy of the **I-Care** program. The findings will offer essential insights for future dementia care programs by using artificial intelligence; and build a strong foundation for scaling **I-Care** programs. By cooperating with Rutgers Health's COPSA Institute for Alzheimer's Disease and Related Disorders, the **I-Care** program can be used in different clinical settings, such as

Care2Caregivers program and Memory Disorders Clinic. The **I-Care** program will be beneficial to quality of life and health wellbeing among dementia caregivers across New Jersey.

Project Title: Development of an **Intelligence Care (I-Care)** Program for Dementia Caregivers

Aim: To develop an artificial **Intelligence Care (I-Care)** program for dementia caregivers

Hypothesis: **I-Care** program will help dementia caregivers manage behavioral and psychological symptoms of dementia and improve self-perceived competence in dementia care and well-being.

Goals: Improving care knowledge, coping skills, and well-being of dementia caregivers in N.J.

Background. In the U.S., approximately 6.7 million people aged 65 and older have Alzheimer's or dementia. By 2060, the number is projected to be 14 million.¹ N.J. has the sixth highest rate of Alzheimer's in the nation, including 185,000 individuals aged 65+ (12% of N.J.'s population).² Caring for people with dementia is highly stressful for family caregivers, as they provide direct care and assistance with activities of daily living, as well as manage behavioral and psychological symptoms of dementia, such as aggression, irritability, abnormal motor behavior, and hallucination.^{3,4} Due to a lack of supportive resources and knowledge of dementia, caregivers may have difficulty managing behavioral symptoms or accessing community resources.⁵ As dementia progresses, caregivers experience increasing strain and burnout, which may result in psychiatric and physical illness.⁵ A retrospective examination of Rutgers' Care2Caregivers program in N.J. reported that dementia caregivers' well-being calls increased yearly. Caregivers' concerns about their well-being include anxiety, depression, loss, grief, posttraumatic stress, trauma, sleep issues, loneliness/isolation, and anger.⁶ In addition, studies report that after caring for people with dementia, many caregivers experience frustration, anger, disappointment, or guilt.^{7,8}

Social-media based intervention has demonstrated efficacy in improving psychosocial well-being in dementia caregivers.⁹ However, most existing tools rely on static educational

materials or rule-based chatbots that cannot adapt to caregivers' individual situations or emotional states.¹⁰ Generative artificial intelligence (GenAI) driven by large language models such as, ChatGPT introduces innovated possibilities for mapping needs, context-sensitive, emotional, and real-time caregiver support.¹¹ ChatGPT and other similar models can process natural language, recall context within a session, and generate coherent, contextually relevant, and evidence-based caregiving responses to a wide range of prompts and questions. When linked with validated educational resources, a GPT-based system can serve as 24/7 virtual companion, improving care efficiency, improving communication, reducing cost, and providing patient personalized, interactive learning experiences.¹² Despite these advances, applications of GenAI in dementia caregiving remain limited¹³ and largely untested. This project addresses this gap by designing and implementing an Intelligence Care (I-Care) program that merges Rutgers' Care2Caregivers educational framework with GPT-based conversational intelligence to deliver personalized, trustworthy, and accessible support for New Jersey's dementia caregiver population.

Project Design and Methods. Step 1: Curriculum and Scenario Design. We will establish an advisory panel, including geriatric psychiatry/family nurse practitioners and medical doctors, Care2Caregivers specialists, and computer science experts to develop educational modules and conversational prompts. Content will include the education materials of Care2Caregivers program¹⁴ with the themes focusing on introduction to dementia, being a caregiver for dementia, managing symptoms of dementia and dealing with behavior changes, self-care with stress coping techniques, and social support and available resources. [These modules will be translated into structured knowledge bases with retrieval-augmented generation \(RAG\) integration to support evidence-based and context-aware model responses.](#) Step 2: System Development (I-Care Platform). The I-Care prototype will be [implemented as a multi-agent conversational system](#)

powered by a GPT-based reasoning engine. The platform will dynamically retrieve and synthesize content from the Care2Caregivers knowledge base via RAG, ensuring accuracy, contextual relevance, and consistency. Within the multi-agent framework, specialized agents will handle distinct functions, including an Education Agent for structured learning delivery, an Emotion-Support Agent for empathetic dialogue and stress monitoring, and a Coordinator Agent for managing dialogue flow and maintaining contextual continuity. The system will feature (1) natural-language and voice interfaces for accessibility, and (2) context-aware dialogue management for tracking caregiver stress patterns and emotional well-being over time. Step 3: Useability Testing and Refinement. After **I-Care** program is developed, we will test it among 6 dementia caregivers. Participants will be invited to a conference room and instructed to download **I-Care** program onto their smart phone. Both quantitative usability metrics (System Usability Scale) and qualitative feedback (ease of use, trust in AI responses, emotional tone) will guide iterative refinements.

Resources and Facilities Available. Educational guides and brochures from the Care2Caregivers program, published by Rutgers Health's Comprehensive Services on Aging (COPSA) Institute for Alzheimer's Disease and Related Disorders¹² are available for curriculum development. The Care2Caregivers Program coordinator agrees to refer dementia caregivers to test and refine the **I-Care** resources tool; and further test the efficacy of the **I-Care** program (see support letter).

Evaluation Methodology. We will conduct a mixed method usability evaluation to assess the functionality, user's experience, and perceived usefulness of the **I-Care** prototype developed in Step 3. We aim to recruit 6 dementia caregivers from the Care2Caregivers program. Inclusion criteria for caregivers include: (1) being at least 18 years old and having the ability to speak and read English; (2) providing care to an individual with dementia for a minimum of four hours per

day. Caregivers will be excluded if they have cognitive impairment or have been the primary caregiver for less than three months. Participants will be invited to download and explore the I-Care program on their smartphones. They will complete a brief demographic survey and participate in a guided usability session, during which they will perform structured tasks such as searching educational materials, engaging in chatbot dialogue, and navigating stress-management resources. Researchers will observe navigation behavior, note system errors, and record time-to-task completion. Usability will be assessed using the System Usability Scale¹⁵ (SUS) and a short satisfaction questionnaire evaluating accuracy, clarity, empathy of AI responses, and perceived trust. Descriptive statistics will summarize SUS scores and satisfaction ratings. A focus group interview will be conducted immediately following the usability session to collect in-depth feedback on user experience, emotional tone, and perceived value of the GenAI component. Findings will guide iterative revisions of the **I-Care** interface, AI conversational prompts, and educational content. The final refined version will form the basis for future pilot feasibility testing in a larger caregiver cohort.

Possible Outcomes and Future Implications. Caregiving for people with dementia poses several challenges that may seriously compromise quality of life. After using **I-Care** program, caregivers are expected to better understand the behavioral and psychological symptoms of dementia, improve their self-efficacy of dementia care, improve their coping skills and well-being. By offering real-time educational resources and interactive support, the **I-Care** program has the potential to reduce caregiver burden and emotional distress, while also lowering healthcare costs by decreasing unnecessary clinic visits and hospitalizations. For research implication, this grant will be used as leverage to obtain future funding to test the feasibility and efficacy of the **I-Care** program. Building on this pilot, the project will provide important preliminary data for larger

studies supported by NIH or other large institutes. The findings will offer essential insights for future dementia care programs by using artificial intelligence; and build a strong foundation for scaling **I-Care** programs. For clinical implications, the COPSA Institute for Alzheimer's Disease and Related Disorders is a statewide program that provides trained staff and consultation to community members throughout New Jersey. This innovative resource tool will provide accessible, trustworthy, and continuous support for caregivers and may serve as a scalable model for improving dementia care and caregiver well-being across New Jersey.

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